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f = @(x) sin(x);
a = 0; b = pi;
n = 8;
ep = 1e-6;

fprintf('exacta          : %.10f\n', 2);
fprintf('--- prob 1 ---\n');
fprintf('trapez (n=%d) : %.10f\n', n, trap(f,a,b,n));
fprintf('drept (n=%d) : %.10f\n', n, drept(f,a,b,n));
fprintf('simp (n=%d) : %.10f\n', n, simp(f,a,b,n));
fprintf('--- prob 3 ---\n');
fprintf('atrap          : %.10f\n', atrap(f,a,b,ep));
fprintf('adrecht        : %.10f\n', adrecht(f,a,b,ep));
fprintf('asimp          : %.10f\n', asimp(f,a,b,ep));
fprintf('--- prob 4 ---\n');
[q4, ~] = romberg(f,a,b,ep);
fprintf('romberg        : %.10f\n', q4);
fprintf('--- prob 5 ---\n');
fprintf('adquad         : %.10f\n', adquad(f,a,b,ep));

%pb1

function q = trap(f, a, b, n)
% formula repetata a trapezului
    h = (b-a)/n;
    s = 0;
    for k = 1:n-1
        s = s + f(a + k*h);
    end
    q = (h/2) * (f(a) + f(b) + 2*s);
end

function q = drept(f, a, b, n)
% formula repetata a dreptunghiurilor (mijloc)
    h = (b-a)/n;
    s = 0;
    for k = 1:n
        s = s + f(a + (k-1)*h + h/2);
    end
    q = h * s;
end

function q = simp(f, a, b, n)
% formula repetata a lui simpson (n trebuie par)
    if mod(n,2) ~= 0
        n = n + 1;
    end
    h = (b-a)/n;
    sp = 0;
    si = 0;
    for k = 1:n-1
        xk = a + k*h;

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        if mod(k,2) == 0
            sp = sp + f(xk);
        else
            si = si + f(xk);
        end
    end
    q = (h/3) * (f(a) + f(b) + 2*sp + 4*si);
end

%pb3

function q = atrap(f, a, b, tol)
% adaptiv trapez
    q = steptrap(f, a, b, tol, 4);
end

function q = steptrap(f, a, b, tol, m)
    q1 = trap(f, a, b, m);
    q2 = trap(f, a, b, 2*m);
    if abs(q1 - q2) < tol
        q = q2;
    else
        c = (a+b)/2;
        q = steptrap(f,a,c,tol,m) + steptrap(f,c,b,tol,m);
    end
end

function q = adrept(f, a, b, tol)
% adaptiv dreptunghi
    q = stepdrept(f, a, b, tol, 4);
end

function q = stepdrept(f, a, b, tol, m)
    q1 = drept(f, a, b, m);
    q2 = drept(f, a, b, 2*m);
    if abs(q1 - q2) < tol
        q = q2;
    else
        c = (a+b)/2;
        q = stepdrept(f,a,c,tol,m) + stepdrept(f,c,b,tol,m);
    end
end

function q = asimp(f, a, b, tol)
% adaptiv simpson
    q = stepsimp(f, a, b, tol, 4);
end

function q = stepsimp(f, a, b, tol, m)
    q1 = simp(f, a, b, m);
    q2 = simp(f, a, b, 2*m);
    if abs(q1 - q2) < tol
        q = q2;
    else

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        c = (a+b)/2;
        q = stepsimp(f,a,c,tol,m) + stepsimp(f,c,b,tol,m);
    end
end

%pb4

function [q, R] = romberg(f, a, b, tol)
% metoda romberg: trapeze + extrapolare richardson
    nmax = 20;
    R = zeros(nmax, nmax);
    h = b - a;
    R(1,1) = (h/2) * (f(a) + f(b));
    q = R(1,1);
    for k = 2:nmax
        h = (b-a) / 2^(k-1);
        s = 0;
        for i = 1:2^(k-2)
            s = s + f(a + (i - 0.5)*2*h);
        end
        R(k,1) = 0.5*R(k-1,1) + h*s;
        for j = 2:k
            p = 4^(j-1);
            R(k,j) = (p*R(k,j-1) - R(k-1,j-1)) / (p-1);
        end
        if abs(R(k,k) - R(k-1,k-1)) < tol
            q = R(k,k);
            R = R(1:k, 1:k);
            return;
        end
    end
    q = R(nmax,nmax);
    R = R(1:nmax, 1:nmax);
end

%pb5

function q = adquad(f, a, b, ep)
% cuadratura adaptiva simpson + extrapolare (alg 1)
    c = (a+b)/2;
    fa = f(a); fb = f(b); fc = f(c);
    q = qstep(f, a, b, ep, fa, fc, fb);
end

function q = qstep(f, a, b, ep, fa, fc, fb)
% pas recursiv al adquad
    h = b - a;
    c = (a+b)/2;
    fd = f((a+c)/2);
    fe = f((c+b)/2);
    q1 = (h/6) * (fa + 4*fc + fb);
    q2 = (h/12) * (fa + 4*fd + 2*fc + 4*fe + fb);
    if abs(q1 - q2) < ep
        q = q2 + (q2 - q1)/15;
    end
end

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    else
        qa = qstep(f, a, c, ep, fa, fd, fc);
        qb = qstep(f, c, b, ep, fc, fe, fb);
        q  = qa + qb;
    end
end

exacta      : 2.0000000000
--- prob 1 ---
trapez (n=8) : 1.9742316019
drept  (n=8) : 2.0129090856
simp   (n=8) : 2.0002691699
--- prob 3 ---
atrap      : 1.9999929145
adrept     : 2.0000047237
asimp      : 2.0000000645
--- prob 4 ---
romberg    : 2.0000000000
--- prob 5 ---
adquad     : 1.9999999997
```

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